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# Spin in the Neurosurgical trauma literature: prevalence and associated factors – a systematic review protocol



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**KEYWORDS**

Spin; Traumatic brain injury; Neurosurgery; Metascience

**WORD COUNT**

1628 words

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# Spin in the Neurosurgical trauma literature: prevalence and associated factors – a systematic review protocol


## ABSTRACT

Introduction: Spin is defined as an inaccurate interpretation of results, intentionally or not, leading to equivocal conclusions and misdirecting readers to look at the data in an overly optimistic way. Previous studies have shown a high prevalence of spin in scientific papers and this systematic review aims to investigate the nature and prevalence of spin in the neurosurgical trauma literature. Any associated factors will be identified to guide future research practice recommendations.

Methods and analysis: The Preferred Reporting Item for Systematic review and Meta-Analyses (PRISMA) recommendations will be followed. Randomized clinical trials (RCT) that enrolled only traumatic brain injury (TBI) patients and investigated any type of intervention (surgical or non-surgical) will be eligible for inclusion. The MEDLINE / PubMed database will be searched for articles in English published in fifteen top-ranked journals. Spin will be defined as (1) a focus on statistically significant results not based on the primary outcome; (2) interpreting statistically nonsignificant results for a superiority analysis of the primary outcome; (3) Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results; (4) Conclusion focused in the per-protocol or as-treated analysis instead of the intention to treat (ITT) results; (5) Incorrect statistical analysis; (6) Republication of a significant secondary analysis without proper acknowledgment of the primary outcome analysis result. Traditional descriptive statistics will be used to present RCT characteristics. Standardized differences between the groups with or without spin will be calculated as proposed by Yang et al. The variables with a standardized difference equal or above 0.2 and 0.5 will be considered weakly and strongly associated with spin respectively.

Ethics and dissemination: This study will not involve primary data collection and patients will not be involved.


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Resgistration: Registrated on Open Science Framework. Registration DOI 10.17605/OSF.IO/H3FGY.

**Key words**: Spin; Traumatic brain injury; Neurosurgery; Metascience

## STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is the first systematic review to evaluate the prevalence of spin in the neurosurgical trauma literature.
- Inclusion of studies with different interventions will allow identification of good and poor examples of randomized clinical trials.
- The inclusion of only randomized clinical trials may make our results more conservative and strength.
- Exclusion of papers not published in specific journals may mean that important additional findings are missed.

## INTRODUCTION

Spin is defined as an inaccurate interpretation of results, intentionally or not, leading to equivocal conclusions and misdirecting readers to look at the data in an overly optimistic way (1). Spin is considered by many as a consequence of the highly competitive and commercial academic system all scientists work within, since it puts negative findings in a more palatable way to editors, journals, patients, funders and readers (2).

Spin may be present in the results, discussion or conclusion sections. Boutron et al. demonstrated in a series of randomized clinical trials (RCT) that spin was present in 29,2%, 43,1% and 50,0% of the studies at the results, discussion and conclusion sections respectively (3). Series of observational studies exhibited more than 80% of spin prevalence (4). Such findings represent a serious problem for the scientific community.

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There are three main different spin strategies. First, to focus on statistically significant results not based on the primary outcome, as a within-group comparisons, secondary outcomes or subgroup analyses. Second, interpreting statistically nonsignificant results for a superiority analysis of the primary outcome as treatment equivalence or comparable effectiveness. Finally, claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results (3).

Spin comes in different shapes and sizes, and medical fields with a less robust, fragile scientific output may be more biased by such malpractice. Considering that the quantity and quality of the neurosurgical literature remains suboptimal, it would be thought-provoking to acknowledge how often spins are encountered, how they are presented and its potential to the specialty practice (5–8).

Therefore, this systematic review aims to investigate the nature and prevalence of spin in the neurosurgical trauma literature and identify any associated factors which could guide future good research practice recommendations.

## METHODS AND ANALYSIS

### Protocol and registration

This systematic review and meta-analysis will be reported following the Preferred Reporting Item for Systematic review and Meta-Analyses (PRISMA) guidelines.

### Eligibility criteria

Published RCT that enrolled only trauma brain injury (TBI) patients and investigated any type of intervention (surgical or non-surgical) will be eligible for inclusion. Studies without a clear primary outcome will be excluded. The following journals will be selected for screening based in the impact factor and importance to the neurosurgical trauma literature: New England Journal of Medicine (NEJM), Lancet, Lancet Neurology, Journal of the American Medical Association (JAMA), JAMA Neurology, Journal of Neurology, Neurosurgery and Psychiatry (JNNP), Neurosurgery, Journal of Neurosurgery, Neurosurgical Focus, World Neurosurgery,



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Acta Neurochirurgica, Journal of Neurotrauma, Intensive Care Medicine, Critical Care, and Neurocritical Care.

**Information sources and search strategy**

The MEDLINE / PubMed database will be searched for articles in English published from January 1960 to July 2020. The descriptors (((((Traumatic brain injury[Title/Abstract]) OR (TBI[Title/Abstract])) OR (Brain trauma[Title/Abstract])) OR (Brain concussion[Title/Abstract])) OR (Brain contusion[Title/Abstract])) OR (head trauma[Title/Abstract]) OR (head injury[Title/Abstract]) OR (brain injury[Title/Abstract]) will be used. The filters "Clinical trial" and "Randomized clinical trials" will be applied.

**Study selection**

The search strategy aimed to achieve a sample of RCTs published in the neurosurgical trauma literature. Spin in sections other than the conclusion (manuscript or abstract), such as the results and discussion sections, will not be considered due to a perceived higher subjectivity in their definition. After all, a reasonable presentation of results not based on the primary outcome, and their discussion is not necessarily a bad practice - and indeed expected. All articles' titles and abstracts will be screened by two authors (A and J) for eligibility. The selected articles will be adjudicated by a third author (DJFS) and disagreements will be resolved by consensus. Additional studies identified in the reference section of the selected articles could be included if the eligibility criteria will be fulfilled.

**Data collection and analysis**

Data will be abstracted and recorded on a standardized form regarding: journal, year of publication, study country (high income or low-and-middle income), first and last authors affiliations (neurosurgery or other), the presence of a statistician among the authors, single-center or multicenter trial, type of trial design (superiority, non-inferiority or equivalence), the presence of a priori sample size and power calculation, type of TBI (mild, moderate, severe), setting (pre-hospital or intra-hospital), intervention (surgical, drug, other), allocation concealment and blinding, number of patients, follow-up period, the event rates in the 2 treatment arms, the presence of a post hoc power calculation, the discussion of lack of power



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as a limitation, funding (industry, independent or none) and conflict of interest (when explicitly stated). The full text of the included articles will be systematically reviewed by two authors (A and J). The interobserver agreement and the $\kappa$ statistic will be calculated. Disagreements will be resolved by a third author (DJFS). The data analysis will be blinded to the authors and institutions of the study. GRADE (Grading of Recommendations, Assessment, Development and Evaluations) framework will be used to assess the level of evidence of each paper, varying from very low up to high.

Spin will be defined as (1) a focus on statistically significant results not based on the primary outcome, as a within-group comparisons, secondary outcomes or subgroup analyses; (2) interpreting statistically nonsignificant results for a superiority analysis of the primary outcome as treatment equivalence or comparable effectiveness; (3) Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results; (4) Conclusion focused in the per-protocol or as-treated analysis instead of the intention to treat (ITT) results; (5) Incorrect statistical analysis; (6) Republication of a significant secondary analysis without proper acknowledgment of the primary outcome analysis result.

Traditional descriptive statistics will be used to present the included RCT characteristics. Standardized differences between the groups with or without spin will be calculated as proposed by Yang et al. (9) The variables with a standardized difference equal or above 0.2 and 0.5 will be considered weakly and strongly associated with spin respectively. All analyses will be conducted with the SPSS software (IBM Corp. SPSS Statistics for Windows, version 24.0. Armonk, NY)

**Risk of bias in individual studies**

The entire text of each included paper was evaluated in a structured fashion for prespecified attributes relating to the assessment and ultimate claim of equivalency. Cochrane's tool to assess the risk of bias in randomized trials was used (10). This way, the investigators were able to access power and sample size calculation, allocation concealment, and many other criteria regarding the methodological quality of each RCT.

**Patient and public involvement**



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No patient involved

## ETHICS AND DISSEMINATION

This study will not involve primary data collection and patients will not be involved. Therefore, formal ethical approval will not be required. The final systematic review will be published in a peer-reviewed journal and presented at appropriate conferences. This protocol may be adapted for the analysis of other innovative surgical and invasive procedures.

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## AUTHORS' CONTRIBUTIONS

DJFS conceived the research idea. JVMPO, ALFOJR and DJFS designed and drafted the study protocol. AGK and WSP all made critical review of the study protocol and edited the final manuscript

## FUNDING

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## COMPETING INTERESTS

The authors declare that they have no competing interests



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# PRISMA-P 2015 Checklist

**This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al:** Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - **Moher D, Stewart L & Shekelle P:** Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

| Section/topic | # | Checklist item | Information reported | | Line number(s) |
|---|---|---|---|---|---|
| | | | Yes | No | |
| **ADMINISTRATIVE INFORMATION** | | | | | |
| **Title** | | | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | X | | 2 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | ☐ | ☐ | |
| **Registration** | 2 | If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract | X | | ABSTRACT |
| **Authors** | | | | | |
| Contact | 3a | Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author | X | | |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | X | | 174 |
| **Amendments** | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | ☐ | ☐ | |
| **Support** | | | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | X | | 165 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | X | | 165 |
| Role of sponsor/funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | X | | 165 |
| **INTRODUCTION** | | | | | |
| **Rationale** | 6 | Describe the rationale for the review in the context of what is already known | X | | 45 |
| | 7 | Provide an explicit statement of the question(s) the review will address with reference to | X | | 66 |

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| Section/topic | # | Checklist item | Information reported | | Line number(s) |
|---|---|---|---|---|---|
| | | | Yes | No | |
| **Objectives** | | participants, interventions, comparators, and outcomes (PICO) | | | |
| **METHODS** | | | | | |
| **Eligibility criteria** | 8 | Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review | X | | 74 |
| **Information sources** | 9 | Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage | X | | 84 |
| **Search strategy** | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | X | | 84 |
| ***STUDY RECORDS*** | | | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | X | | 101 |
| Selection process | 11b | State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis) | X | | 101 |
| Data collection process | 11c | Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | X | | 101 |
| **Data items** | 12 | List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications | X | | 102 |
| **Outcomes and prioritization** | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | X | | 117 |
| **Risk of bias in individual studies** | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | X | | 101 |
| ***DATA*** | | | | | |
| **Synthesis** | 15a | Describe criteria under which study data will be quantitatively synthesized | X | | 101 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., $I^2$, Kendall's tau) | X | | 101 |
| | 15c | Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression) | X | | 101 |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | X | | 101 |
| **Meta-bias(es)** | 16 | Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies) | | X | |

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| Section/topic | # | Checklist item | Information reported | | Line number(s) |
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| | | | Yes | No | |
| **Confidence in cumulative evidence** | 17 | Describe how the strength of the body of evidence will be assessed (e.g., GRADE) | X | | 114 |

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| Primary Subject Heading: | Neurology |
| Secondary Subject Heading: | Surgery, Medical publishing and peer review, Epidemiology |
| Keywords: | STATISTICS & RESEARCH METHODS, NEUROSURGERY, NEUROLOGY |
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# TITLE PAGE

## Spin in the Neurosurgical trauma literature: prevalence and associated factors – a systematic review protocol


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**KEYWORDS**

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**WORD COUNT**

2162 words

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Resgistration: Registrated on Open Science Framework. Registration DOI 10.17605/OSF.IO/H3FGY.

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- This is the first systematic review to evaluate the prevalence of spin in the neurosurgical trauma literature.
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## INTRODUCTION

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There are three main different spin strategies. First, to focus on statistically significant results not based on the primary outcome, as a within-group comparisons, secondary outcomes or subgroup analyses. Second, interpreting statistically nonsignificant results for a superiority analysis of the primary outcome as treatment equivalence or comparable effectiveness. Finally, claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results (3).

Spin comes in different shapes and sizes, and medical fields with a less robust, fragile scientific output may be more biased by such malpractice. Considering that the quantity and quality of the neurosurgical literature remains suboptimal, it would be thought-provoking to acknowledge how often spins are encountered, how they are presented and its potential to the specialty practice (5–8).

Therefore, this systematic review aims to investigate the nature and prevalence of spin in the neurosurgical trauma literature and identify any associated factors which could guide future good research practice recommendations.

## METHODS AND ANALYSIS

### Protocol and registration

This systematic review and meta-analysis will be reported following the Preferred Reporting Item for Systematic review and Meta-Analyses (PRISMA) guidelines.

### Eligibility criteria

Published RCT that enrolled only trauma brain injury (TBI) patients and investigated any type of intervention (surgical or non-surgical) will be eligible for inclusion. Studies without a clear primary outcome will be excluded. The following journals will be selected for screening based in the impact factor and importance to the neurosurgical trauma literature: New England Journal of Medicine (NEJM), Lancet, Lancet Neurology, Journal of the American Medical Association (JAMA), JAMA Neurology, Journal of Neurology, Neurosurgery and Psychiatry (JNNP), Neurosurgery, Journal of Neurosurgery, Neurosurgical Focus, World Neurosurgery,



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Acta Neurochirurgica, Journal of Neurotrauma, Intensive Care Medicine, Critical Care, Neurocritical Care, Journal of Trauma and Acute Care Surgery or Critical Care Medicine.

**Information sources and search strategy**

The MEDLINE / PubMed database will be searched for articles in English published from January 1960 to July 2020. The descriptors ((((((((((Traumatic brain injury[Title/Abstract]) OR (TBI[Title/Abstract])) OR (Brain trauma[Title/Abstract])) OR (Brain concussion[Title/Abstract])) OR (Brain contusion[Title/Abstract])) OR (Head trauma[Title/Abstract])) OR (Head injury[Title/Abstract])) OR (Brain injury[Title/Abstract])) OR (Traumatic subarachnoid hemorrhage[Title/Abstract])) OR (subdural hematoma[Title/Abstract])) OR (epidural hematoma[Title/Abstract])) OR (traumatic intraparenchymal hemorrhage[Title/Abstract]) will be used. The filters “Clinical trial” and “Randomized clinical trials” will be applied.

**Study selection**

The search strategy aimed to achieve a sample of RCTs published in the neurosurgical trauma literature. Spin in sections other than the conclusion (manuscript or abstract), such as the results and discussion sections, will not be considered due to a perceived higher subjectivity in their definition. After all, a reasonable presentation of results not based on the primary outcome, and their discussion is not necessarily a bad practice - and indeed expected. All articles’ titles and abstracts will be screened by two authors (A and J) for eligibility. The selected articles will be adjudicated by a third author (DJFS) and disagreements will be resolved by consensus. Additional studies identified in the reference section of the selected articles could be included if the eligibility criteria will be fulfilled.

**Data collection and analysis**

Data will be abstracted and recorded on a standardized form regarding: journal, year of publication, study country (high income or low-and-middle income), first and last authors affiliations (neurosurgery or other), the presence of a statistician among the authors, single-center or multicenter trial, type of trial design (superiority, non-inferiority or equivalence), the presence of a priori sample size and power calculation, type of TBI (mild, moderate, severe), setting (pre-hospital or intra-hospital), intervention (surgical, drug, other), allocation



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concealment and blinding, number of patients, follow-up period, the event rates in the 2 treatment arms, the presence of a post hoc power calculation, the discussion of lack of power as a limitation, funding (industry, independent or none) and conflict of interest (when explicitly stated). The full text of the included articles will be systematically reviewed by two authors (A and J). The interobserver agreement and the $\kappa$ statistic will be calculated. Disagreements will be resolved by a third author (DJFS). The data analysis will be blinded to the authors and institutions of the study. GRADE (Grading of Recommendations, Assessment, Development and Evaluations) framework will be used to assess the level of evidence of each paper, varying from very low up to high.

Spin will be defined as (1) a focus on statistically significant results not based on the primary outcome, as a within-group comparisons, secondary outcomes or subgroup analyses; (2) interpreting statistically nonsignificant results for a superiority analysis of the primary outcome as treatment equivalence or comparable effectiveness; (3) Claiming or emphasizing the beneficial effect of the treatment despite statistically nonsignificant results; (4) Conclusion focused in the per-protocol or as-treated analysis instead of the intention to treat (ITT) results; (5) Incorrect statistical analysis; (6) Republication of a significant secondary analysis without proper acknowledgment of the primary outcome analysis result.

Traditional descriptive statistics will be used to present the included RCT characteristics. Standardized differences between the groups with or without spin will be calculated as proposed by Yang et al. They are indexes which measure the effect size between two groups (9). The variables with a standardized difference equal or above 0.2 and 0.5 will be considered weakly and strongly associated with spin respectively. All analyses will be conducted with the SPSS software (IBM Corp. SPSS Statistics for Windows, version 24.0. Armonk, NY)

**Risk of bias in individual studies**

The entire text of each included paper will be evaluated in a structured fashion for prespecified attributes relating to the assessment and ultimate claim of equivalency. Cochrane's tool to assess the risk of bias in randomized trials was used (10). This way, the investigators were



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able to access power and sample size calculation, allocation concealment, and many other criteria regarding the methodological quality of each RCT.

**Patient and public involvement**

No patient involved

## ETHICS AND DISSEMINATION

This study will not involve primary data collection and patients will not be involved. Therefore, formal ethical approval will not be required. The final systematic review will be published in a peer-reviewed journal and presented at appropriate conferences. This protocol may be adapted for the analysis of other innovative surgical and invasive procedures.

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## AUTHORS' CONTRIBUTIONS

DJFS conceived the research idea. JVMPO, ALFOJR and DJFS designed and drafted the study protocol. AGK and WSP all made critical review of the study protocol and edited the final manuscript

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## COMPETING INTERESTS

The authors declare that they have no competing interests



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# PRISMA-P 2015 Checklist

This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - Moher D, Stewart L & Shekelle P: Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

| Section/topic | # | Checklist item | Information reported | | Line number(s) |
|---|---|---|---|---|---|
| | | | Yes | No | |
| **ADMINISTRATIVE INFORMATION** | | | | | |
| **Title** | | | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | X | | 2 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | ☐ | ☐ | |
| **Registration** | 2 | If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract | X | | ABSTRACT |
| **Authors** | | | | | |
| Contact | 3a | Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author | X | | |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | X | | 174 |
| **Amendments** | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | ☐ | ☐ | |
| **Support** | | | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | X | | 165 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | X | | 165 |
| Role of sponsor/funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | X | | 165 |
| **INTRODUCTION** | | | | | |
| **Rationale** | 6 | Describe the rationale for the review in the context of what is already known | X | | 45 |

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| Section/topic | # | Checklist item | Information reported | | Line number(s) |
|---|---|---|---|---|---|
| | | | Yes | No | |
| **Objectives** | 7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | X | | 66 |
| **METHODS** | | | | | |
| **Eligibility criteria** | 8 | Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review | X | | 74 |
| **Information sources** | 9 | Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage | X | | 84 |
| **Search strategy** | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | X | | 84 |
| ***STUDY RECORDS*** | | | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | X | | 101 |
| Selection process | 11b | State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis) | X | | 101 |
| Data collection process | 11c | Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | X | | 101 |
| **Data items** | 12 | List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications | X | | 102 |
| **Outcomes and prioritization** | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | X | | 117 |
| **Risk of bias in individual studies** | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | X | | 101 |
| ***DATA*** | | | | | |
| **Synthesis** | 15a | Describe criteria under which study data will be quantitatively synthesized | X | | 101 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., $I^2$, Kendall's tau) | X | | 101 |
| | 15c | Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta- | X | | 101 |

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| Section/topic | # | Checklist item | Information reported | | Line number(s) |
|---|---|---|---|---|---|
| | | | Yes | No | |
| | | regression) | | | |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | X | | 101 |
| **Meta-bias(es)** | 16 | Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies) | | X | |
| **Confidence in cumulative evidence** | 17 | Describe how the strength of the body of evidence will be assessed (e.g., GRADE) | X | | 114 |

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